

MSDM 5003 Lecture 11

13 November 2025

Correlations and Power Laws

Outline:

1. Do markets follow random walks?

2. Stylized facts of markets

3. Long-time correlations

Detrended fluctuation analysis (DFA)

Power spectral density (PSD)

4. Power laws and exponential laws

5. Stock price return distribution

6. Market index return distribution

Differentiating power laws and exponential laws

1. Do Markets Follow Random Walks?

(Econophysics Chapter 2)

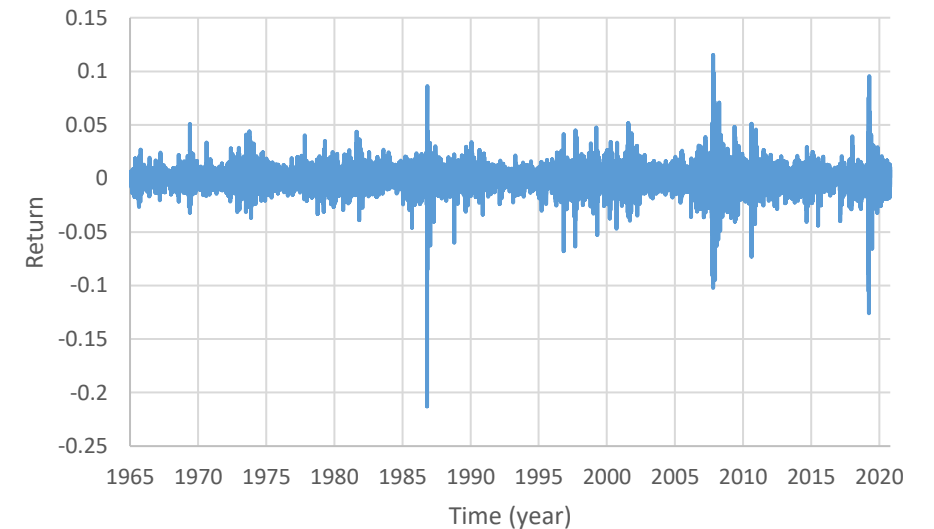
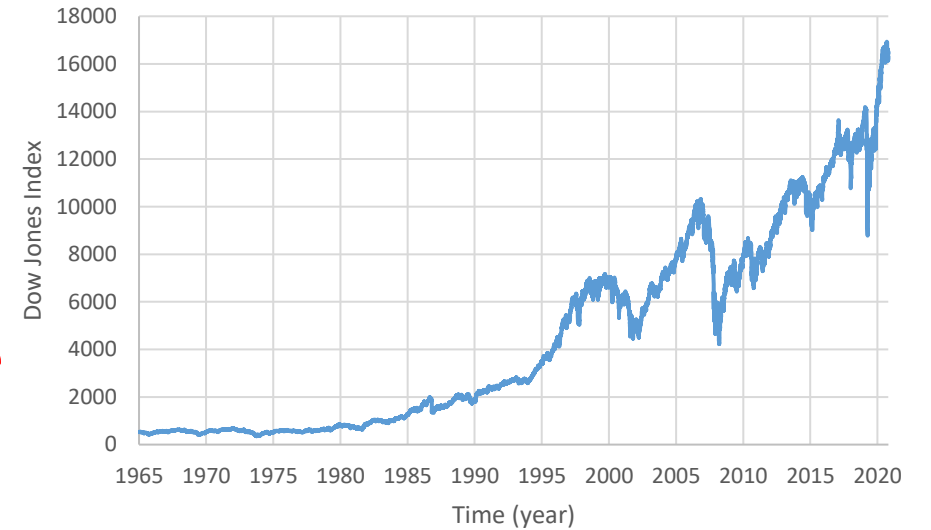
Bachelier (1900) first proposed that market price movements can be described by random walks.

However, if the price of a stock follows a random walk, it would be impossible to predict future price movements.

Let $P(t)$ be the price of a stock at time t .

Return:

$$r_{\tau}(t) = \ln P(t + \tau) - \ln P(t).$$



Comparison with Gaussian Distribution

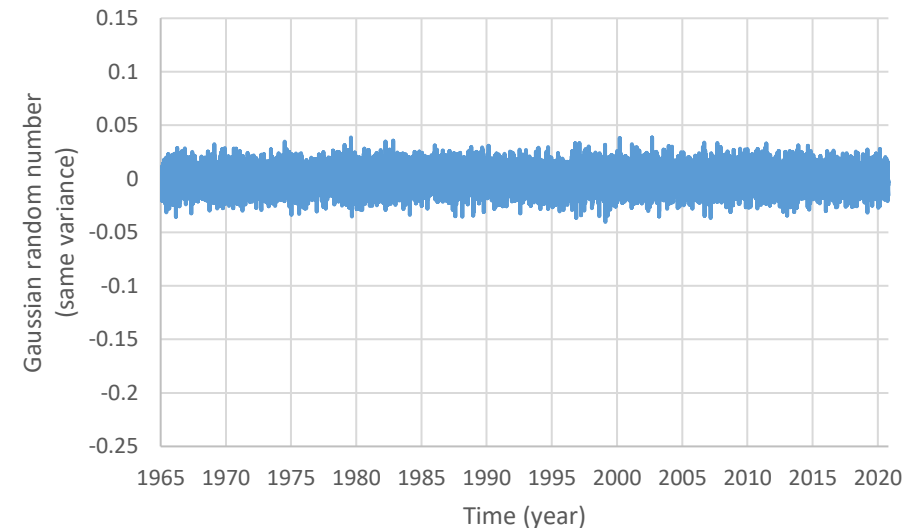
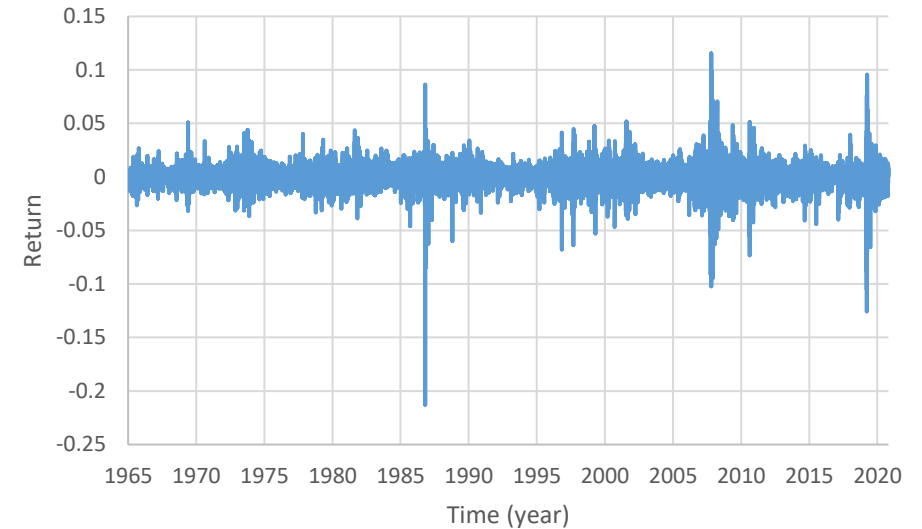
The steady-state distribution of a random walk is a Gaussian (normal) distribution:

$$P(r) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{r^2}{2\sigma^2}\right).$$

The probability of large fluctuations is very low. Bachelier (1900) first proposed that market price movements can be described by random walks.

However, the return distribution of real financial data looks very different from the Gaussian distribution.

There are occasional large fluctuations.



Financial Data Distributions are Fat-Tailed

In 1960s, Mandelbrot pointed out that empirical return distributions are “**fat-tailed**” and also more peaked at the center when compared with Gaussian distribution.

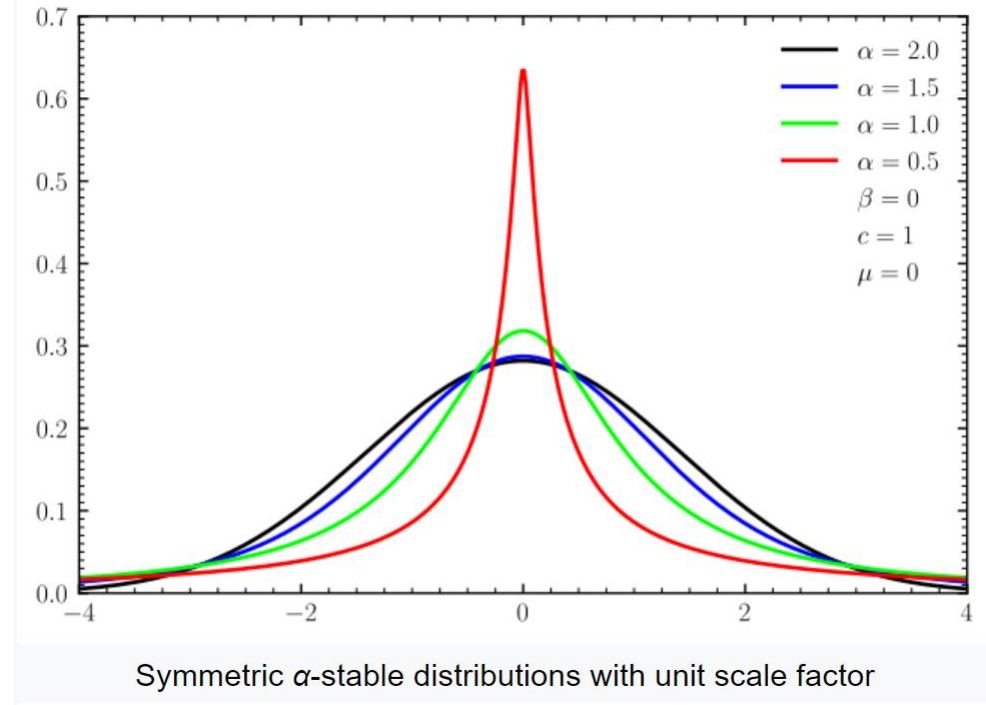
He found that the return distribution was a **Levy-stable distribution**.

The Levy-stable distribution depends on the parameters α, β, c, μ . For financial data, $\alpha \approx 1.7$. Assume that $c = 1$, the tail part is given by a **power law behavior**,

$$P_{\text{Levy}}(|x|) \sim |x|^{-(1+\alpha)}.$$

Since $\alpha < 2$, it has an infinite variance.

$$\text{Variance} \sim \lim_{R \rightarrow \infty} \int_0^R dx P_{\text{Levy}}(|x|) x^2 \sim R^{2-\alpha} \rightarrow \infty.$$



2. Stylized facts of Financial Data

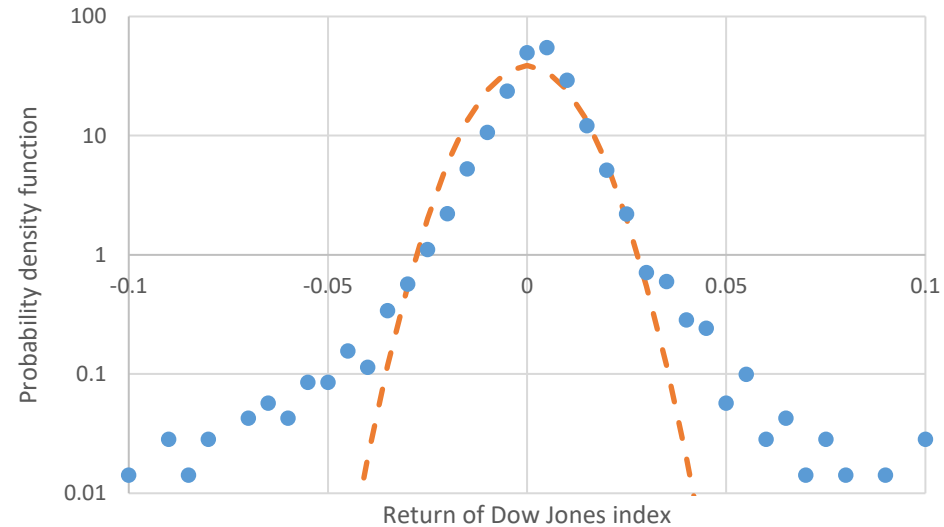
Stylized Facts of Returns (Econophysics Subsection 2.2.2)

Empirical studies of returns show that they are more sophisticated than random walks. They are characterized by “stylized” facts. Here, we mention two of them:

- (1) **Fat tails**: Large returns asymptotically follow a power law $F(r) \sim |r|^{-\alpha}$, with $\alpha > 2$ ($F(r)$ is the cumulative distribution).
- (2) **Volatility clustering**: the volatility of price appears to cluster in time and large fluctuations tend to be followed by large fluctuations and small changes tend to be followed by small ones.

Volatility clustering is characterized by a power law of the autocorrelation function of $|r(t)|$. On the other hand, the autocorrelation of $r(t)$ is short-ranged (e.g. following an exponential law).

Fat Tails



Compared with the Gaussian distribution (orange), the tail of the return distribution **decays much slower**, and the peak at the center is much **sharper**.

Power Laws

In practice, the tail of the power law distribution is often too sparse to be measured accurately.

It is more reliable to measure the **complementary cumulative distribution function** (CCDF) of the power law distribution, which also obeys a power law,

$$P(r > x) \sim \int_x^\infty dr \, r^{-(1+\alpha)} = \frac{x^{-(1+\alpha)+1}}{(1+\alpha)-1} \sim x^{-\alpha}.$$

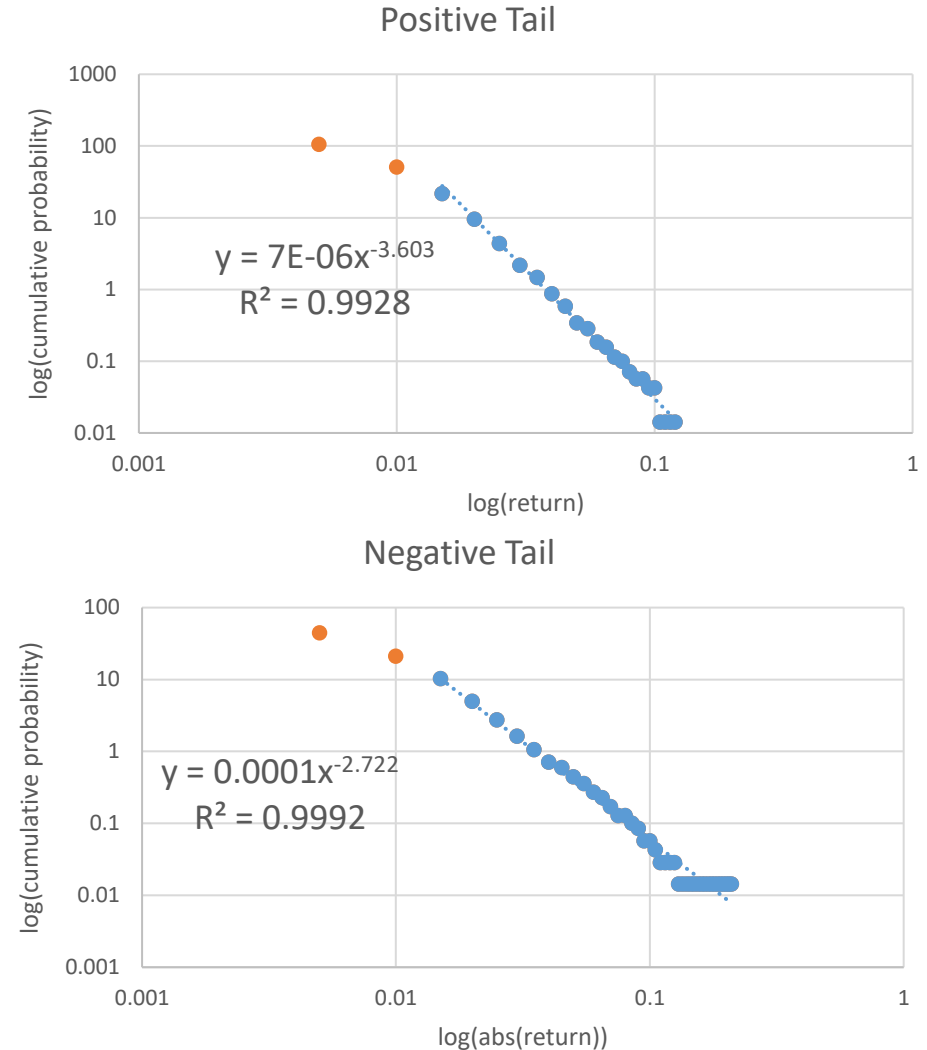
α can be found from the slope of the log-log plot,

$$\ln P(r > x) = -\alpha \ln x + c.$$

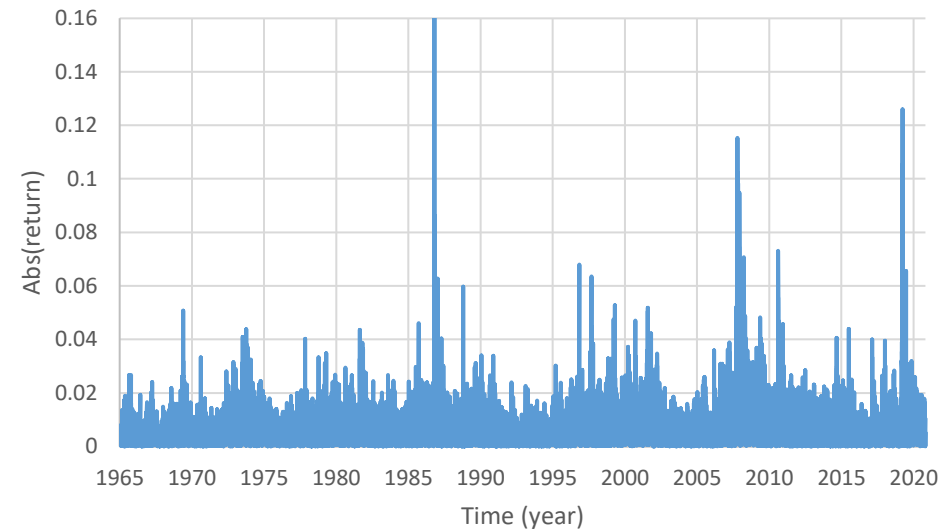
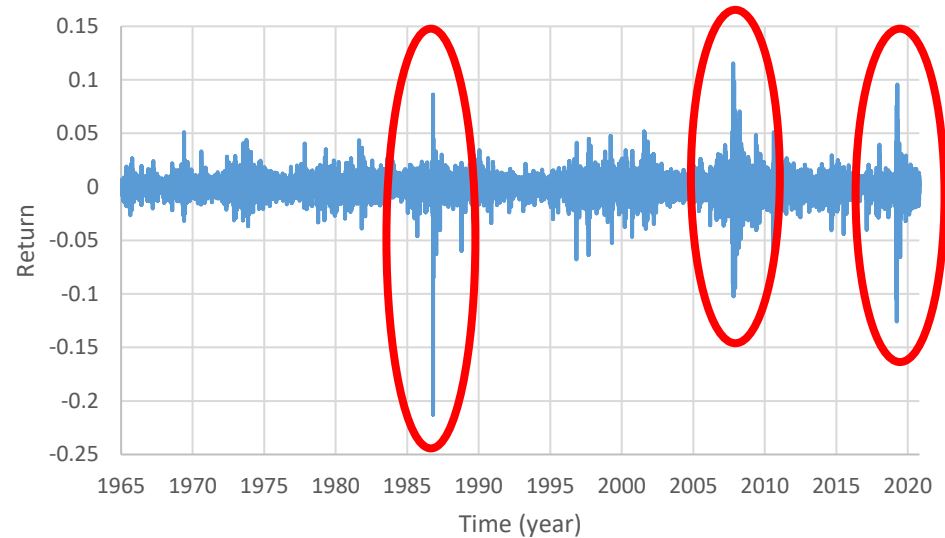
(or other curve fitting functions in Excel etc.)

$\alpha = 3.60$ for positive tail and $\alpha = 2.72$ for negative tail.

Since $\alpha > 2$, the variance is well-defined.

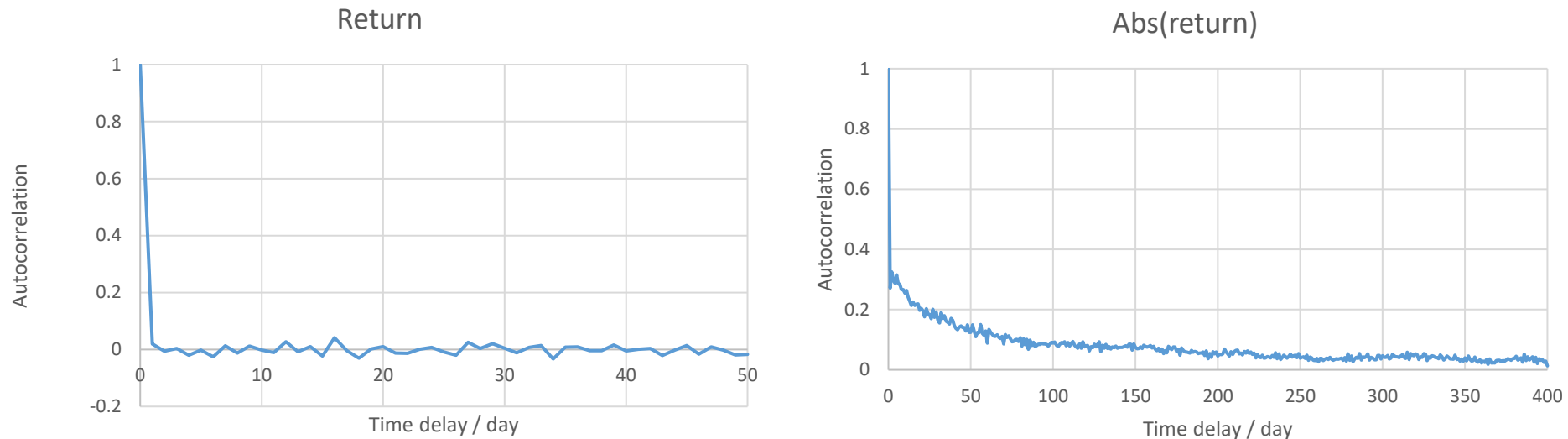


Volatility Clustering



Volatility is measured by the absolute value of the return.
Return and volatility look very different.

Autocorrelation



The autocorrelation of the return at time lag T is defined as

$$C(T) = \frac{\langle [r(t+T) - \langle r \rangle][r(t) - \langle r \rangle] \rangle}{\langle r^2 \rangle - \langle r \rangle^2}.$$

The autocorrelation of the return decays rapidly in time, but the autocorrelation of the volatility decays much slower. This indicates that volatility is clustered.

Figures: Prepared using `plt.acorr`; need to subtract the mean first.

3. Long-time Correlations

Detrended Fluctuation Analysis (DFA)

Consider a time series ξ_i of length T .

Integrate the time series to an unbounded process (like a random walk)

$$X_t = \sum_{i=1}^t (\xi_i - \langle \xi \rangle).$$

Divide the length into N non-overlapping periods of length τ , such that $N\tau = T$.

In each period $i = 1, 2, \dots, N$, **fit** the cumulative sum X_t with $z_{it} = a_i t + b_i$, called the local trend.

Detrend by subtracting the local trend to compute the fluctuation function

$$F(\tau) = \left[\frac{1}{\tau} \sum_{t=(i-1)\tau+1}^{i\tau} (X_t - z_{it})^2 \right]^{\frac{1}{2}}.$$

Compute $F(\tau)$ for different box sizes τ .

DFA¹

Calculate the **correlation exponent** α from the power law relation

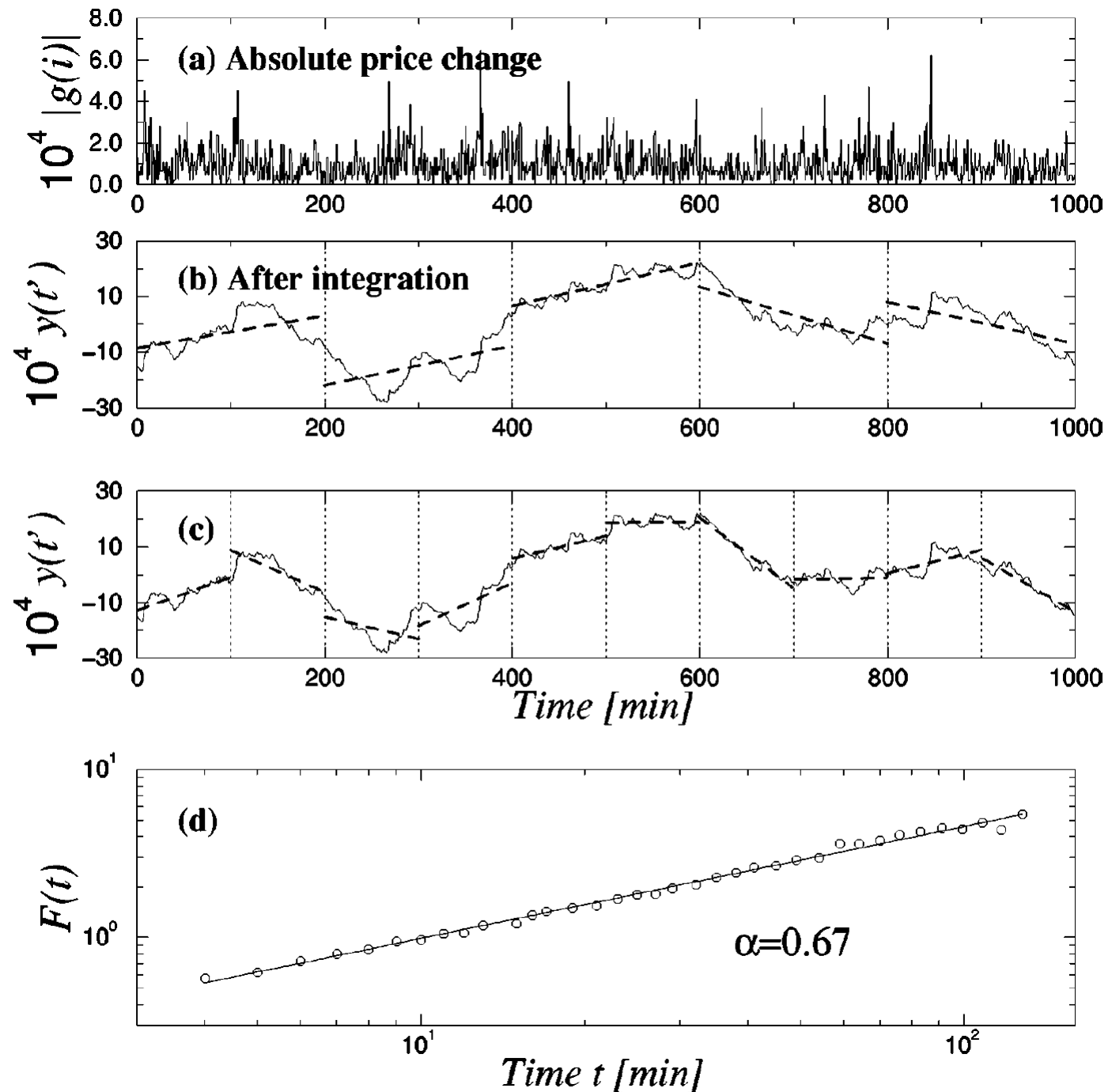
$$F(\tau) \sim \tau^\alpha.$$

If $\alpha = 0.5$, the signal is uncorrelated (white noise).

If $\alpha > 0.5$, there are persistent long-range correlations in the time series.

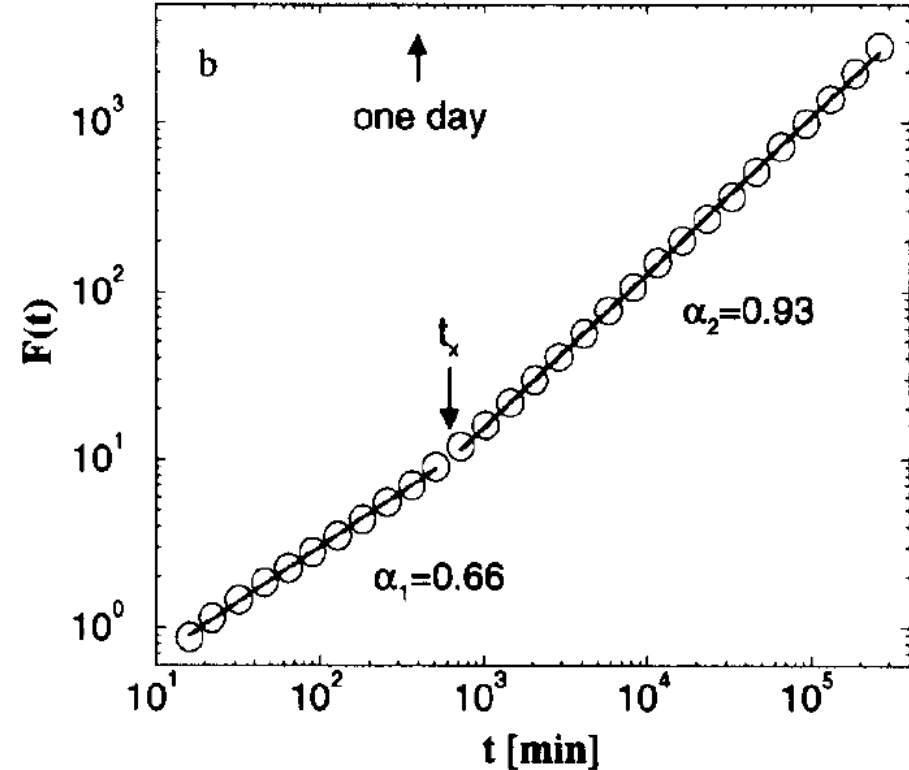
If $\alpha < 0.5$, anti-correlations are present. Large values are likely to be followed by small values and vice versa.

¹Y. Liu, P. Gopikrishnan, P. Cizeau, M. Meyer, C.-K. Peng, and H. E. Stanley, Statistical properties of the volatility of price fluctuations, Phys. Rev. E 60, 1390-1400 (1999).



DFA of Absolute Returns²

- The absolute returns $|r_i|$ of S&P 500 index between January 1984 and December 1996 was analyzed at one-minute intervals.
- $\alpha_1 = 0.66$ at short time scale. $\alpha_2 = 0.93$ at long time scale.
- The crossover occurs at $t_x \approx 600$ min, corresponding to 1.5 trading days.
- This indicates that the short-time and long-time behaviors of the market are different.



²Y. Liu, P. Cizeau, M. Meyer, C.-K. Peng, and H. E. Stanley, Correlations in Economic Time Series, Physica A 245, 437-440 (1997).

Power Spectral Density Analysis²

- The autocorrelation function $C(\tau)$ can be considered as the sum (or integral) of Fourier components $S(f)$ of frequency f .
- $S(f)$ is called the **power spectrum**.
- The low frequency components describes the long-time behavior, and the high frequency components describe the short-time behavior.
- It is calculated by

$$S(f) = \int_{-\infty}^{\infty} d\tau C(\tau) e^{-2\pi i f \tau}.$$

- Often in financial applications, only the amplitude of the power spectrum is studied.

$$|S(f)| = \sqrt{\text{Re}S(f)^2 + \text{Im}S(f)^2},$$

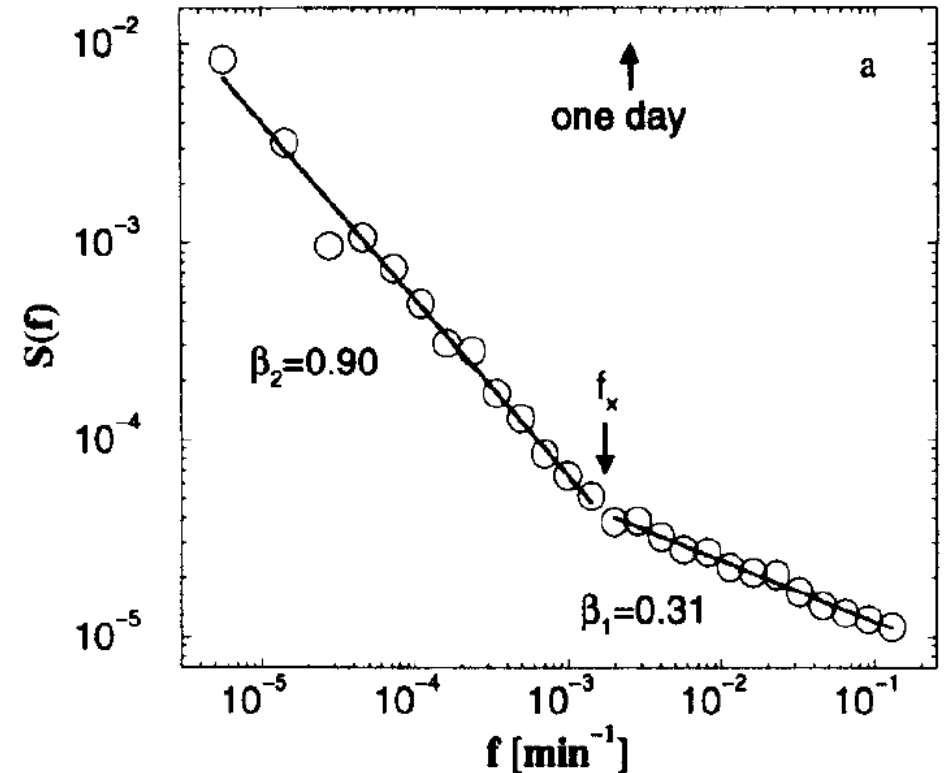
$$\text{where } \text{Re}S(f) = \int_{-\infty}^{\infty} d\tau C(\tau) \cos(2\pi f \tau), \text{Im}S(f) = - \int_{-\infty}^{\infty} d\tau C(\tau) \sin(2\pi f \tau).$$

PSD of Absolute Returns

- The power spectral density of the absolute returns $|r_i|$ follows a power law

$$S(f) \sim \frac{1}{f^\beta}.$$

- $\beta_1 = 0.31$ at high frequency. $\beta_2 = 0.90$ at low frequency.
- The crossover occurs at $f_X \approx \frac{1}{600 \text{ min}}$.
- This is consistent with the result of DFA.



Relation of DFA and PSD Exponents

- The DFA exponent α is related to the PSD exponent β by $\beta = 2\alpha - 1$.
- At short time scale, $\beta_1 = 0.31$, $\alpha_1 = 0.66$, and at long time scale, $\beta_2 = 0.90$, $\alpha_2 = 0.93$.
- Proof: Suppose the correlation function of the time series scales as $C(\tau) \sim \tau^{-\gamma}$.

$$S(f) = \int_{-\infty}^{\infty} d\tau C(\tau) e^{-2\pi i f \tau}.$$

Let $x = f\tau$. Then

$$S(f) \sim f^{-(1-\gamma)} \int_{-\infty}^{\infty} dx x^{-\gamma} e^{-2\pi i x} \Rightarrow \beta = 1 - \gamma.$$

In DFA, note that

$$F(\tau)^2 \sim \langle X(\tau)^2 \rangle = \int_0^\tau dt_1 \int_0^\tau dt_2 \langle \xi(t_1) \xi(t_2) \rangle \sim \int_0^\tau dt_1 \int_0^\tau dt_2 |t_1 - t_2|^{-\gamma} \sim \tau^{2-\gamma} \\ \Rightarrow 2\alpha = 2 - \gamma.$$

Eliminating γ , the result is proved.

4. Power Laws and Exponential Laws

Why care about a power law? (Econophysics Chapter 5)

Distributions with a “typical value” (or “scale”):

Examples:

Gaussian distribution

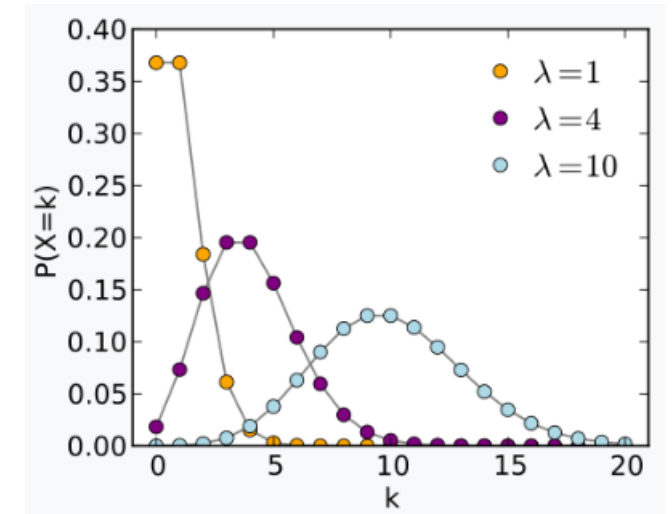
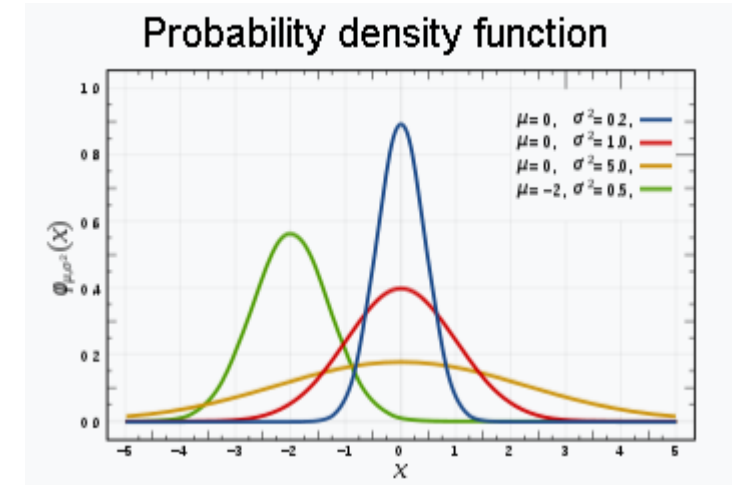
$$N(x) = \frac{e^{-(x-\langle x \rangle)^2/2\sigma^2}}{\sqrt{2\pi\sigma^2}}$$

This distribution is significant only when $x - \langle x \rangle$ is of the order of σ .

Poisson distribution

$$P(n) = \frac{e^{-\lambda}}{n!} \lambda^n$$

This distribution is significant only when $n - \lambda$ is of the order of $\sqrt{\lambda}$.



Power Laws

The exponential distribution is also a short-ranged distribution with the time scale $1/\lambda$,

$$P(t) = \lambda e^{-\lambda t}.$$

However, the power law distribution is not characterized by a length scale:

$$P(x) \propto x^{-k}.$$

Properties:

Absence of typical length scales

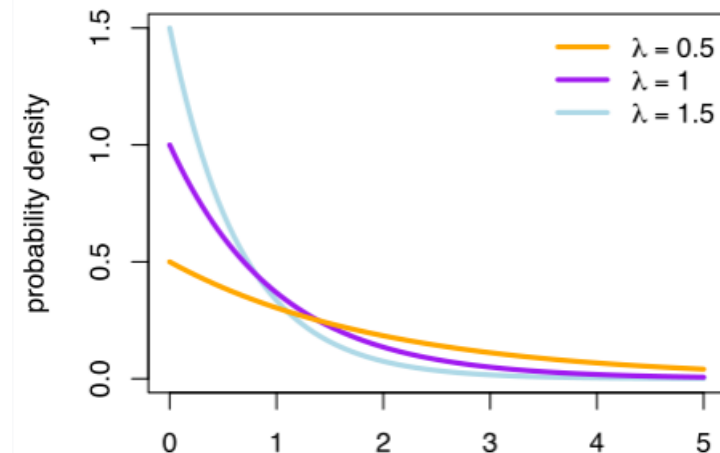
Spanning a wide range of values

Examples:

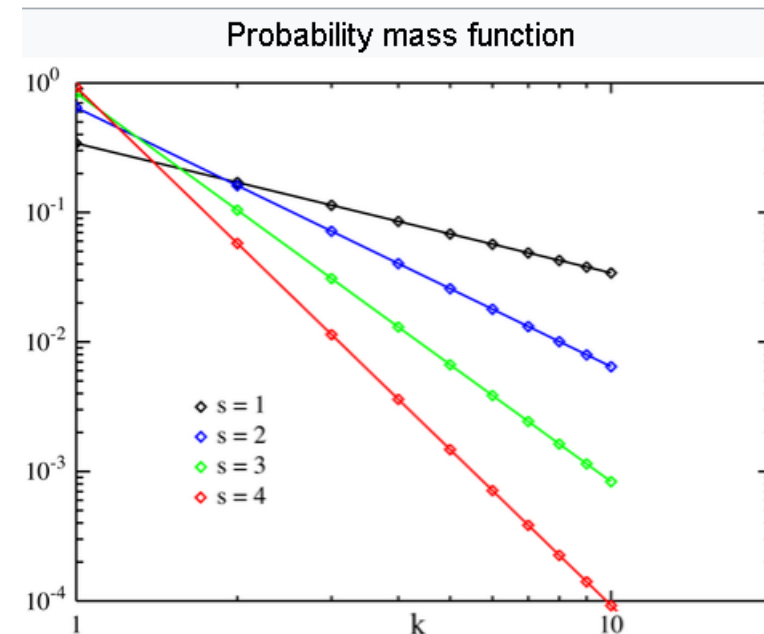
Gravitational force (inverse square law)

Frequency of earthquakes (Gutenberg-Richter law)

Frequency of words in a language (Zipf's law)



Zipf's law



Zipf PMF for $N = 10$ on a log-log scale. The horizontal axis is the index k . (Note that the function is only defined at integer values of k . The connecting lines do not indicate continuity.)

Power Laws in Finance

Consider stock price or market index $p(t)$ at time t .

The return for time-steps of size Δt is given by

$$R(t, \Delta t) \equiv \ln p(t + \Delta t) - \ln p(t) .$$

Note that calculus shows that

$$\ln p(t + \Delta t) - \ln p(t) \approx \frac{[p(t + \Delta t) - p(t)]}{p(t)} ,$$

so that the return is independent of the scale of the price.

How are the returns in a market distributed?

Early theory: random walk model $\Rightarrow$ Gaussian (Bachelier 1900)

Recent theory: interactions in market $\Rightarrow$ power law? Larger fluctuations allowed?

Cumulative Return Distribution

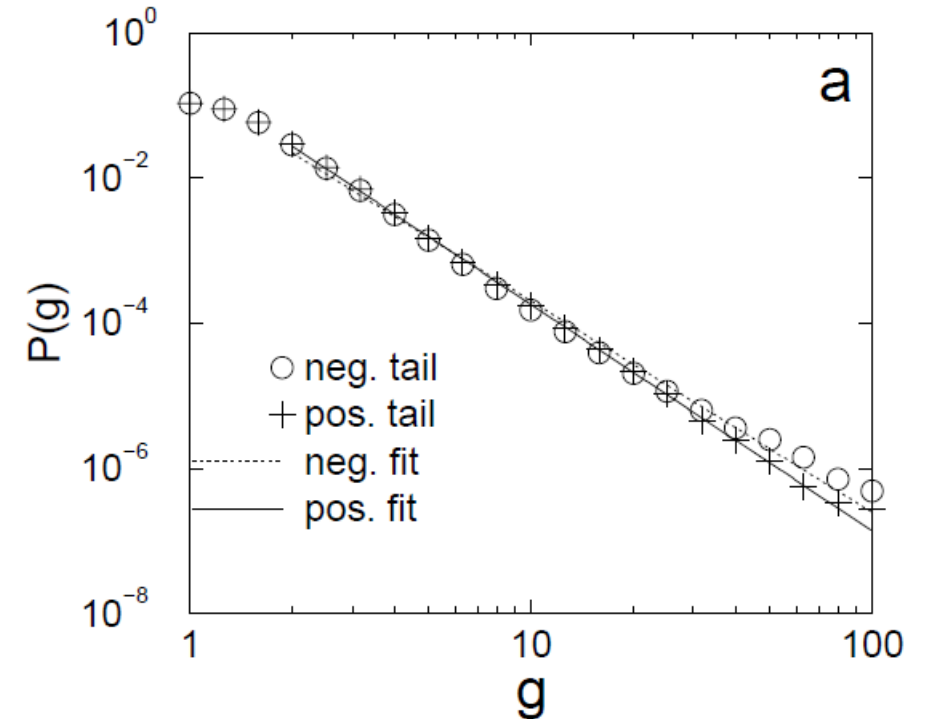
It is more reliable to measure the tail of the cumulative power-law distribution, and the cumulative distribution also obeys a power law.

US stock market database³: $\alpha \approx 3$ (inverse cubic law)

$$P(r > x) \sim x^{-\alpha}$$

Confirmed by studies in S&P 100 stocks,

German stock exchange, London Stock Exchange



³ Gopikrishnan P, Meyer M, Amaral LAN, and Stanley HE (1998) Inverse cubic law for the distribution of stock price variations, *Eur Phys J B* **3**, 139-140.

Controversy

Developed markets: Not all power-laws

Australian stock market: positive tail closer to Gaussian

Hang Seng index: exponential

Nikkei index: exponential

Developing markets: Not all exponentials

Korean stock market: power-law changed to exponential

Some other emerging markets: inverse-cubic law

Criticism: Conclusions from graphical fitting may not be accurate, more reliable techniques required.

5. Stock Price Return Distribution

High-frequency tick-by-tick data from Indian stock market

- (1) Does the market exhibit a price fluctuation different from the inverse cubic law form seen in developed markets?
- (2) If the market is indeed following the inverse cubic law at present, whether this has been converged, starting from an initially different distribution when the market has just begun operation.

(Answer to both in the next few slides: no)

$p_i(t)$ = price of stock i at time t

Price return:

$$R_i(t, \Delta t) \equiv \ln p_i(t + \Delta t) - \ln p_i(t) .$$

To compare the different stocks, the price returns are normalized by dividing them with their volatility $\sigma_i(t)^2 \equiv \langle R_i^2 \rangle - \langle R_i \rangle^2$. Normalized price return:

$$r_i(t, \Delta t) \equiv \frac{R_i(t) - \langle R_i(t) \rangle}{\sigma_i(t)} .$$

Power Law $\alpha = 3$ in Indian Stocks (5 min ret)⁴

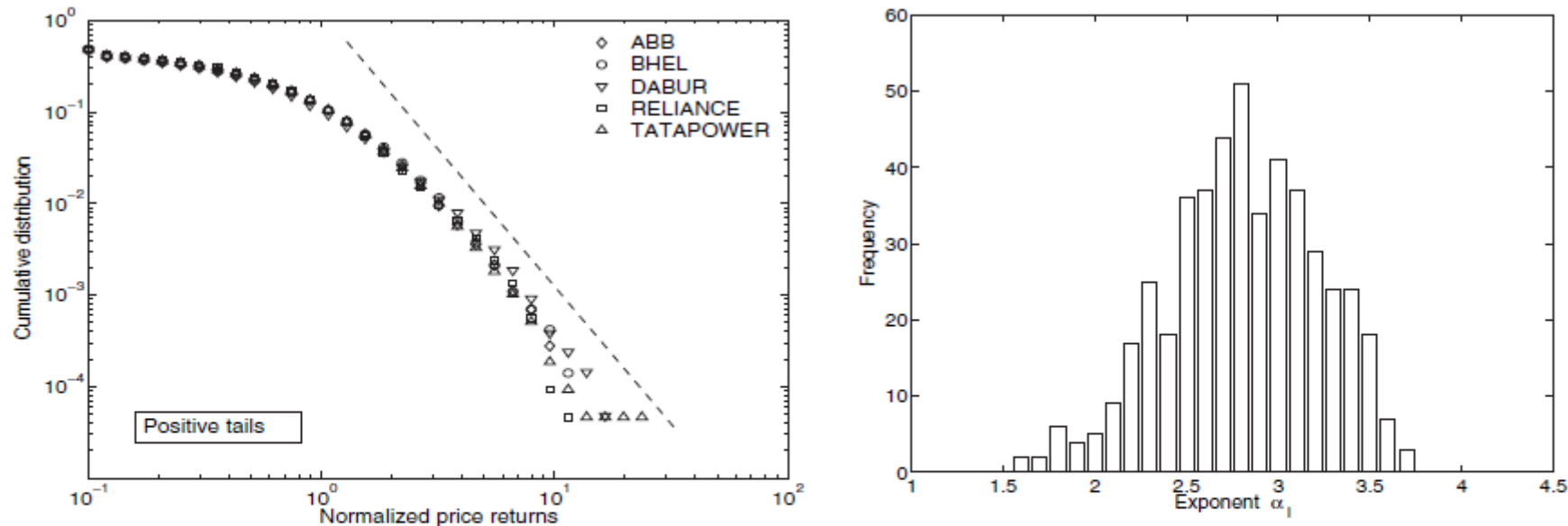


Fig. 1: Left: cumulative distribution of the positive tails of the normalized 5-min returns distribution of 5 stocks chosen arbitrarily from those listed in the NSE for the period January 2003 to March 2004. The broken line indicates a power law with exponent $\alpha = 3$. Right: the histogram of the power law exponents obtained by regression fit for the positive tail of individual cumulative return distributions of 489 stocks. The median of the exponent values is 2.84.

⁴ Pan RK and Sinha S (2007) Self-organization of price fluctuation distribution in evolving markets, *Europhys Lett* **77**, 58004.

Aggregate Data (5 min, 489 stocks)

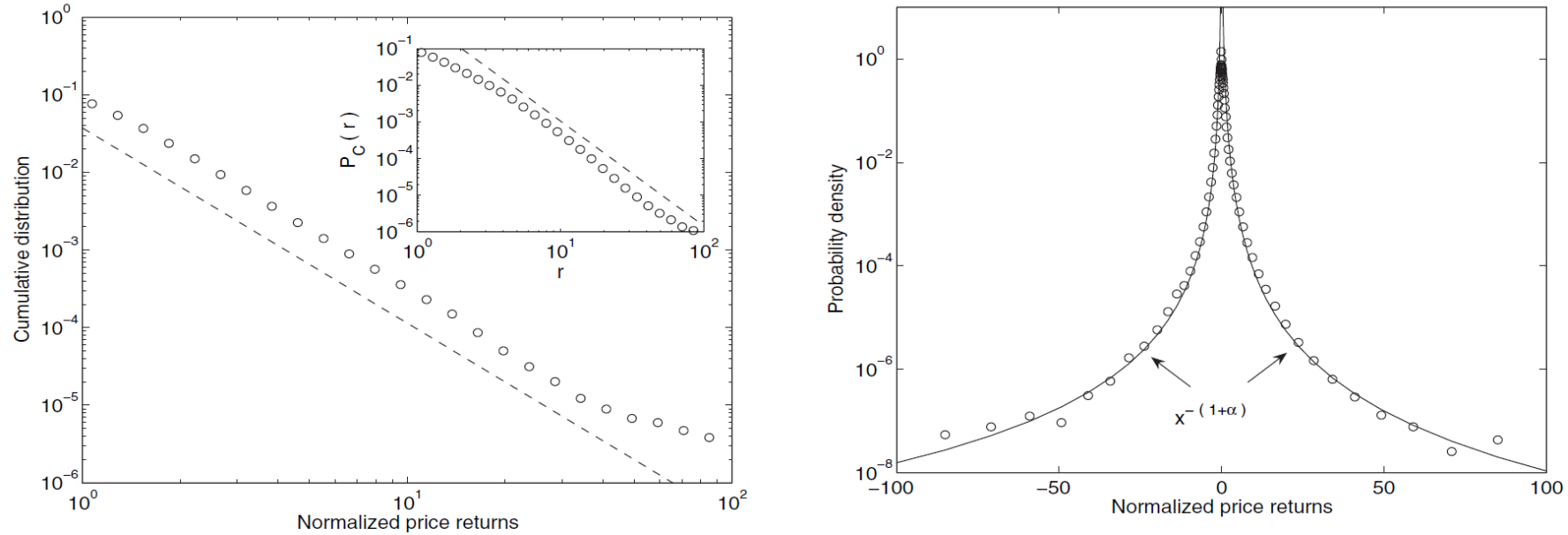


Fig. 2: Left: cumulative distribution of the negative and (inset) positive tails of the normalized returns for the aggregated data of 489 stocks in the NSE for the period January 2003 to March 2004. The broken line for visual guidance indicates the power law asymptotic form. Right: probability density function of the normalized returns. The solid curve is a power law fit in the region 1–50. We find that the corresponding cumulative distribution exponent, $\alpha = 2.87$ for the positive tail and $\alpha = 2.52$ for the negative tail.

$$\alpha = \begin{cases} 2.87 \pm 0.08 & \text{(positive tail)} \\ 2.52 \pm 0.04 & \text{(negative tail)} \end{cases}$$

Hill Estimator of the Exponent

Normalized power-law distribution:

$$P(x) = \frac{\alpha}{x_{\min}} \left(\frac{x}{x_{\min}} \right)^{-\alpha-1} \quad \text{for } x \geq x_{\min}.$$

For n data points, the log likelihood function becomes

$$L(\alpha) = \sum_{i=1}^n \ln \left[\frac{\alpha}{x_{\min}} \left(\frac{x_i}{x_{\min}} \right)^{-\alpha-1} \right].$$

Differentiating with respect to α ,

$$\frac{\partial L(\alpha)}{\partial \alpha} = \sum_{i=1}^n \left[\frac{1}{\alpha} - \ln \left(\frac{x_i}{x_{\min}} \right) \right] = 0.$$

Hence the estimator becomes

$$\hat{\alpha} = \left[\frac{1}{n} \sum_{i=1}^n \ln \left(\frac{x_i}{x_{\min}} \right) \right]^{-1}.$$

Hill Estimator in Practice

In practice, the data points are arranged in descending order such that

$$x_n < \cdots < x_2 < x_1.$$

Based on the largest $k + 1$ values,

$$(x_n < \cdots <) x_{k+1} < \cdots x_2 < x_1,$$

we construct an estimator assuming that x_{k+1} is the cutoff,

$$H_{k,n} = \frac{1}{k} \sum_{i=1}^k \ln \left(\frac{x_i}{x_{k+1}} \right).$$

When $k \rightarrow \infty$ (while $k \ll n$), we expect $H_{k,n} \rightarrow \alpha^{-1}$.

See also J. Nuyts, Inference about the Tail of a Distribution: Improvement on the Hill Estimator, Int. J. Math. and Math. Sci., 924013 (2010).

$\hat{\alpha} = 2.86$ (positive tail) and 2.56 (negative tail).

Dependence on Δt ?

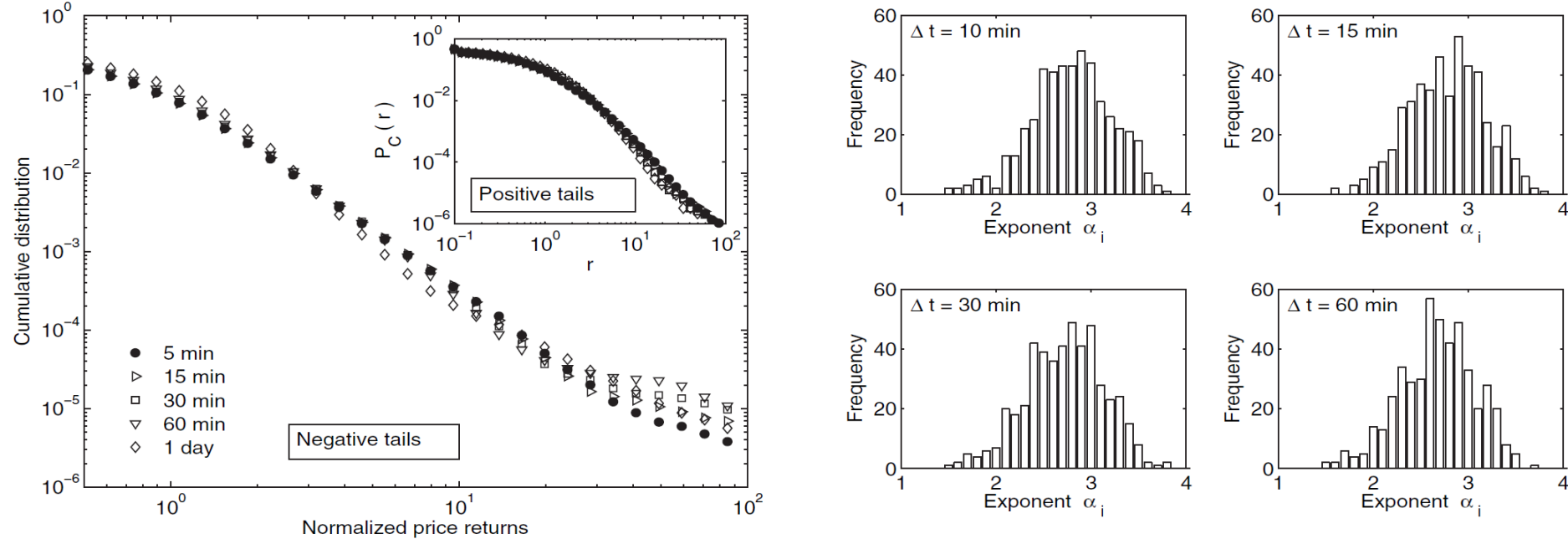


Fig. 3: Left: cumulative distribution of the negative and (inset) positive tails of the normalized returns distribution for different time scales ($\Delta t \leq 1$ day). Right: histograms of the power law exponents for each of the 489 stocks, obtained by regression fit on the positive tail of cumulative return distributions, for different time scales ($10 \text{ min} \leq \Delta t \leq 60 \text{ min}$).

Power-law distribution for Δt up to 1 day. Left: all stocks, right: individual stocks.

Comparison with a Mature Market

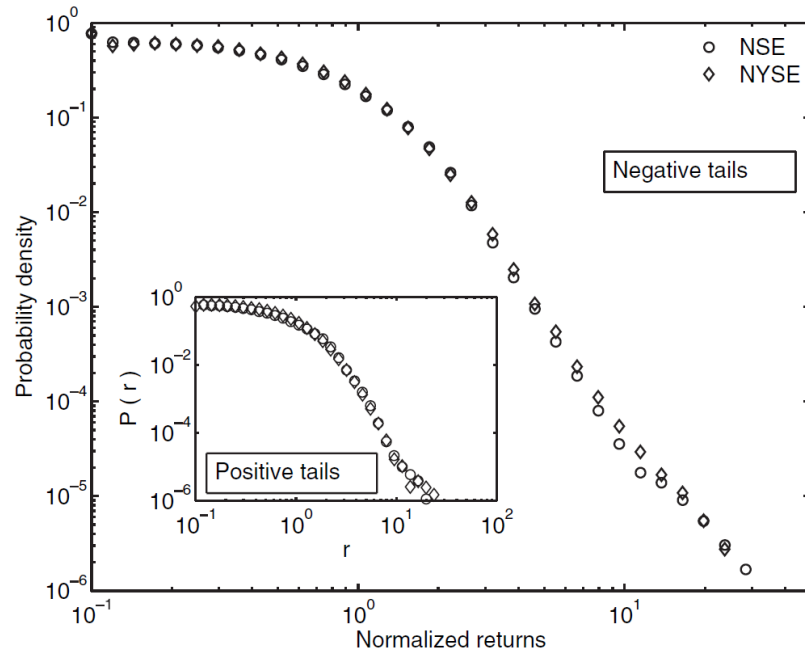


Fig. 4: Comparison of the negative and (inset) positive tails of the normalized daily returns distribution for all stocks traded at NSE ($\circ$) and 500 stocks traded at NYSE ($\diamond$) during the period November 1994 to May 2006.

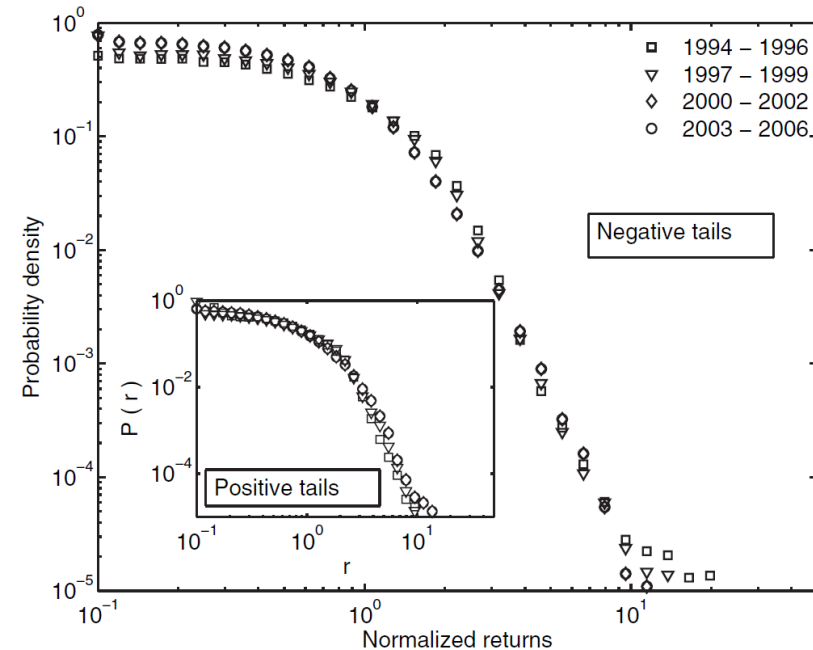


Fig. 5: The negative and (inset) positive tails of the normalized daily returns distribution for all NSE stocks traded during the periods 1994–1996 ($\square$), 1997–1999 (∇), 2000–2002 ($\diamond$) and 2003–2006 ($\circ$).

Left: Similar power laws for NYSE (mature market) and NSE (emerging market).
Right: Similar power laws for different development periods of NSE.
The questions in slide 21 are answered.

Possible Reason for Exponential Law?

Not enough data!

Especially in the tail part!

6. Market Index Return Distribution⁵

Growth of the Indian stock market

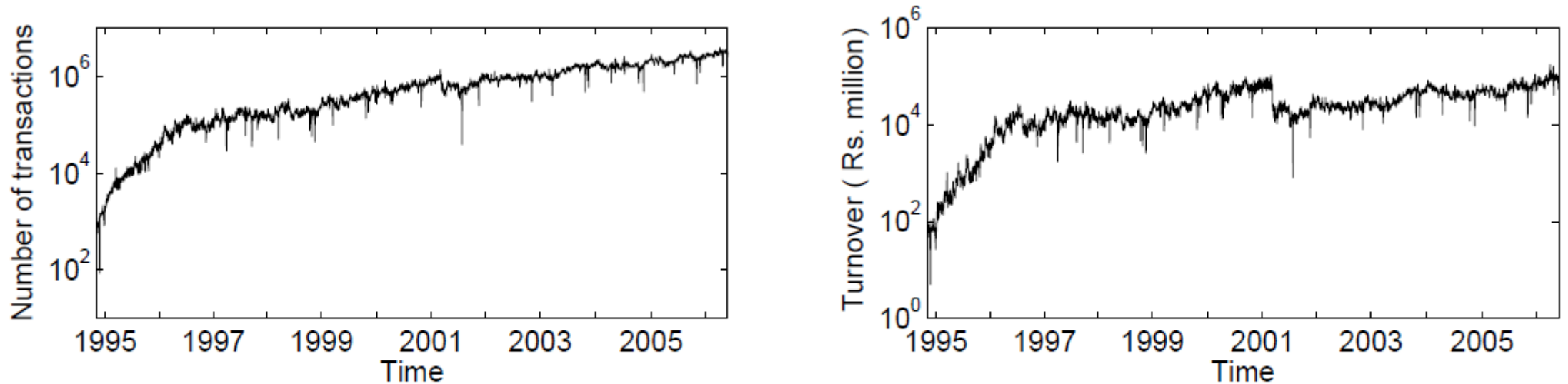


FIG. 1: Time evolution of the National Stock Exchange of India from 1994 – 2006 in terms of (left) the total number of trades and (right) the total turnover (i.e., traded value).

⁵ Pan RK and Sinha S (2008) Inverse cubic law of index fluctuation distribution in Indian markets, *Physica A* **387**, 2055-2065.

Market Index Return

High-frequency data from Indian stock index (NSE), sampled at 1 min intervals

Volatility:

$$\sigma_{\Delta t}^2 \equiv \langle R^2 \rangle - \langle R \rangle^2.$$

Scaled return:

$$r(t, \Delta t) \equiv \frac{R - \langle R \rangle}{\sigma_{\Delta t}(t)}.$$

However, when a single return is very large, it will contribute to the variance as well

⇒ under-estimation of the tail

To avoid this, we remove the contribution of $R(t)$ to itself from the volatility, and the rescaled volatility is defined as

$$\sigma_{\Delta t} = \sqrt{\frac{1}{N-1} \sum_{t' \neq t} \{[R(t', \Delta t)]^2 - \langle R(t', \Delta t) \rangle^2\}}.$$

Remark: The effect of imposing the condition $t' \neq t$ is negligible for Hang Seng index.

Test for the Tail Distribution⁶

TP statistic is used to test whether the tail distribution is Pareto-like (i. e. power law).

TE statistic is used to test whether the tail distribution is exponential.

In TP statistic, consider a power-law distribution with a cut-off u . Let $F(x)$ be the cumulative distribution from $-\infty$ up to u ,

$$1 - F(x) \sim \left(\frac{u}{x}\right)^\alpha \quad \text{for } x \geq u.$$

For a finite sample $x_1, \dots, x_n$, define

$$E_1 = E \left[\ln \frac{X}{u} \right] \quad \text{and} \quad E_2 = E \left[\left(\ln \frac{X}{u} \right)^2 \right].$$

The theoretical values of E_1 and E_2 are

$$E_1 = \int_u^\infty \left(\ln \frac{x}{u} \right) \left(-\frac{d}{dx} \right) \left(\frac{u}{x} \right)^\alpha dx \quad \text{and} \quad E_2 = \int_u^\infty \left(\ln \frac{x}{u} \right)^2 \left(-\frac{d}{dx} \right) \left(\frac{u}{x} \right)^\alpha dx.$$

⁶ Pisarenko V and Sornette D (2006) New statistic for financial return distribution: power law or exponential? *Physica A* **366**, 387-400.

TP Statistic (2)

Let $v = \left(\frac{u}{x}\right)^\alpha$. Then,

$$E_1 = \frac{1}{\alpha} \int_1^0 \ln v \, dv = \frac{1}{\alpha} (v \ln v - v) \Big|_1^0 = \frac{1}{\alpha}.$$

Similarly,

$$E_2 = -\frac{1}{\alpha^2} \int_1^0 (\ln v)^2 \, dv = \frac{1}{\alpha^2} \left(-v(\ln v)^2 \Big|_1^0 + 2 \int_1^0 \ln v \, dv \right) = \frac{2}{\alpha^2}.$$

The TP statistic is defined as

$$\text{TP} = \left[\frac{1}{n} \sum_{k=1}^n \ln \frac{x_k}{u} \right]^2 - \frac{1}{2n} \sum_{k=1}^n \left(\ln \frac{x_k}{u} \right)^2.$$

According to the law of large numbers, if the sample is drawn from the tail of a power law distribution, TP approaches 0 as $n \rightarrow \infty$, irrespective of the value of the exponent α . The standard deviation of TP is of the order of $1/\sqrt{n}$, given by

$$\frac{1}{\sqrt{n}} \times \text{std} \left[2E_1 \ln \frac{x_k}{u} - \frac{1}{2} \left(\ln \frac{x_k}{u} \right)^2 \right] \quad \text{where} \quad E_1 = \frac{1}{n} \sum_{k=1}^n \ln \frac{x_k}{u}.$$

See Ref [6].

TE Statistic (1)

In TE statistic, consider an exponential distribution with a cut-off u . Let $F(x)$ be the cumulative distribution from $-\infty$ up to u ,

$$1 - F(x) \sim \exp\left(-\frac{x - u}{d}\right) \quad \text{for } x \geq u.$$

For a finite sample $x_1, \dots, x_n$, define

$$E_1 = E \left[\ln \left(\frac{X}{u} - 1 \right) \right] \quad \text{and} \quad E_2 = E \left[\left(\ln \left(\frac{X}{u} - 1 \right) \right)^2 \right].$$

The theoretical values of E_1 and E_2 are

$$E_1 = \int_u^\infty \left(\ln \frac{x - u}{u} \right) \left(-\frac{d}{dx} \right) \exp \left(-\frac{x - u}{d} \right) dx,$$

Let $v = \exp \left(-\frac{x - u}{d} \right)$. Then,

$$E_1 = - \int_1^0 \ln \left(\frac{d}{u} \ln \frac{1}{v} \right) dv = \ln \frac{d}{u} - \gamma,$$

$\gamma = 0.577215$ Euler constant. Identity (see Wikipedia): $\gamma = - \int_0^1 \ln \left(\ln \frac{1}{v} \right) dv$.

TE Statistic (2)

$$\begin{aligned} E_2 &= \int_u^\infty \left(\ln \frac{x-u}{u} \right)^2 \left(-\frac{d}{dx} \right) \exp \left(-\frac{x-u}{d} \right) dx \\ &= - \int_1^0 \left(\ln \frac{d}{u} + \ln \left(\ln \frac{1}{v} \right) \right)^2 dv = \left(\ln \frac{d}{u} - \gamma \right)^2 + \frac{\pi^2}{6}. \end{aligned}$$

We have used the identity (see Wikipedia) $\int_0^\infty e^{-x} (\ln x)^2 dx = \gamma^2 + \pi^2/6$.

The TE statistic is defined as

$$\text{TE} = \frac{1}{n} \sum_{k=1}^n \left(\ln \frac{x_k - u}{u} \right)^2 - \left[\frac{1}{n} \sum_{k=1}^n \ln \frac{x_k - u}{u} \right]^2 - \frac{\pi^2}{6}.$$

If the sample is drawn from the tail of an exponential distribution, TE approaches 0 as $n \rightarrow \infty$.
The standard deviation of TE is given by

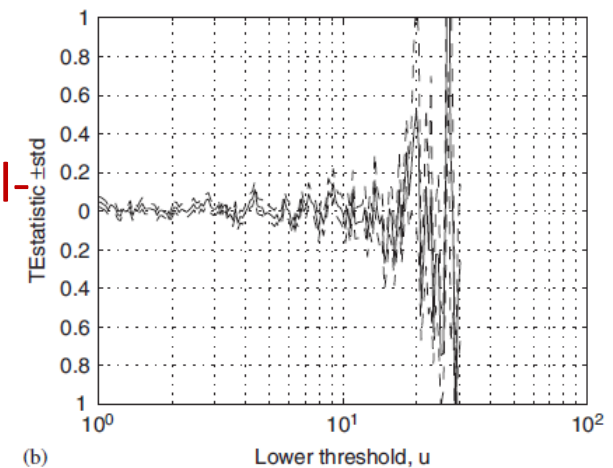
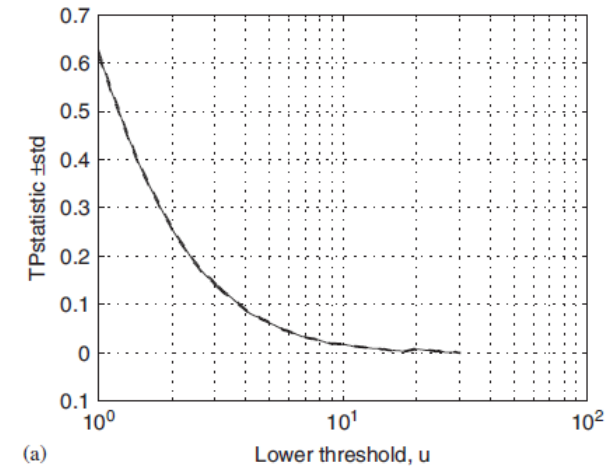
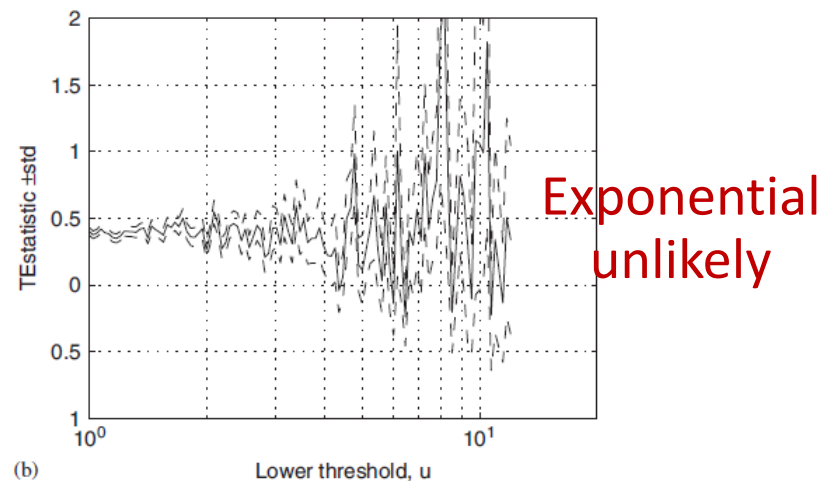
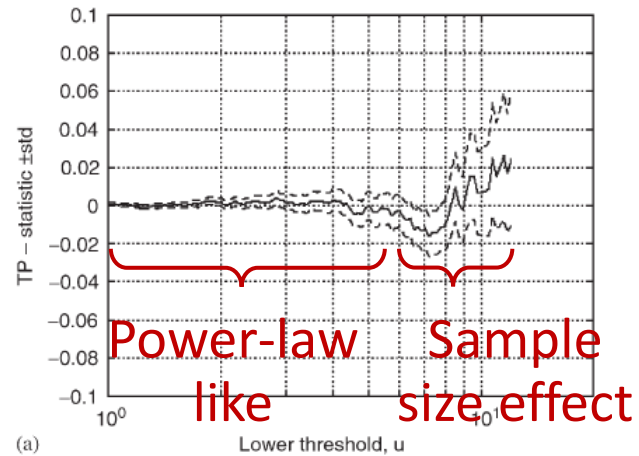
$$\frac{1}{\sqrt{n}} \times \text{std} \left[\left(\ln \frac{x_k - u}{u} \right) - E_1 \right]^2 \quad \text{where} \quad E_1 = \frac{1}{n} \sum_{k=1}^n \ln \frac{x_k - u}{u}.$$

See Ref [6].

Tests on Simulated Data

Simulated Pareto sample

Simulated exponential sample



Exponential-like

Fig. 1. (a) TP -statistic as a function of the lower threshold u , applied to a simulated Pareto sample of size $n = 20,000$ with power index $b = 3$ generated with $u = 1$ as defined in Eq. (1). Increasing u decreases the number of data values used in the calculation of the statistic TP , thus enhancing the fluctuations around 0. The two dashed lines show plus or minus one standard deviation std estimated as exposed in the text. (b) TE -statistic of the same synthetic data.

Fig. 2. (a) TP -statistic as a function of the lower threshold u , applied to a simulated exponential sample of size $n = 20,000$ with form parameter $d = 4$ generated with $u = 1$ as defined in Eq. (8). The two dashed lines show plus or minus one standard deviation std estimated as exposed in the text. (b) TE -statistic of the same synthetic data.

Application to Dow Jones Industrial Average (DJIA)

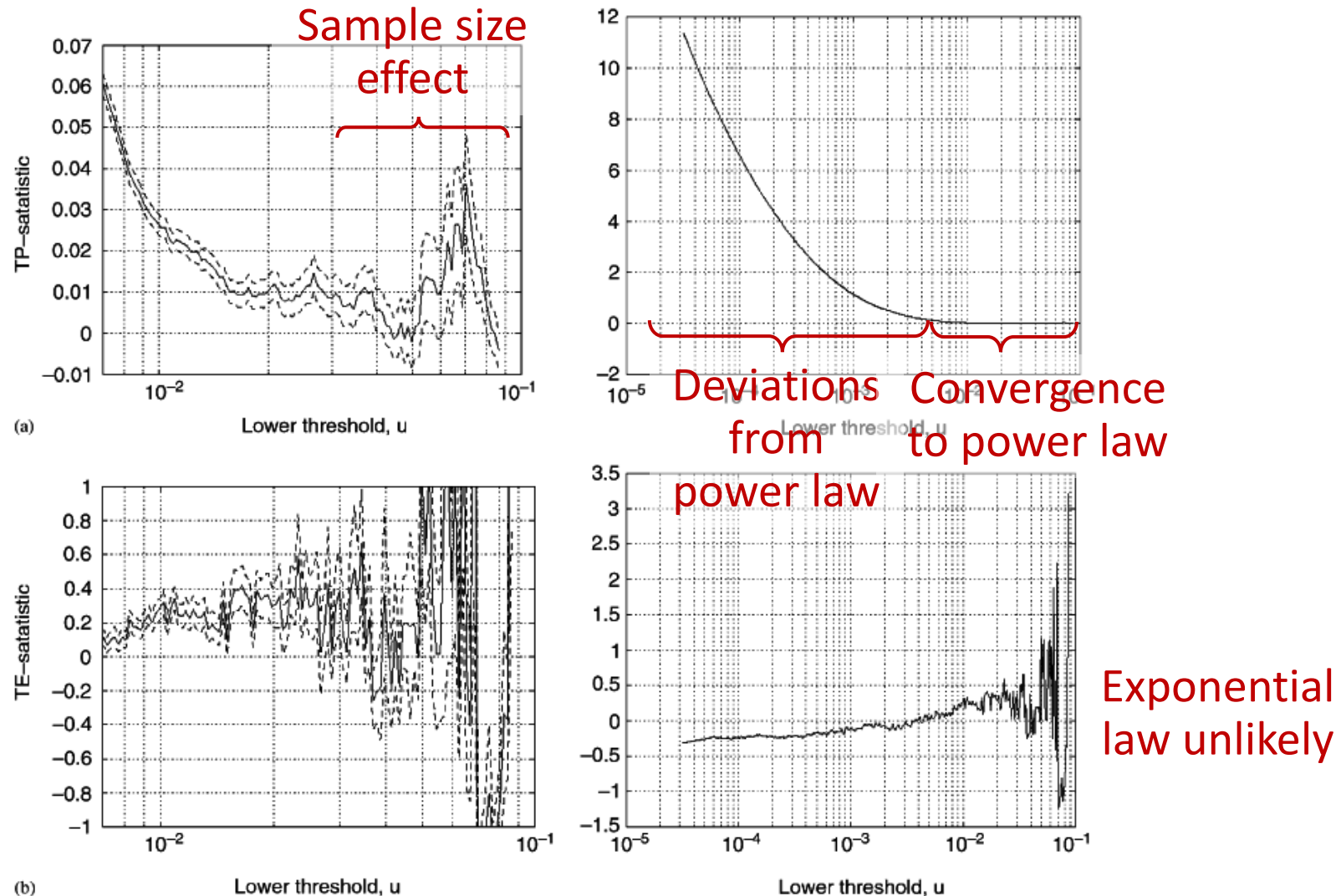


Fig. 6. TP -statistic $TP(u)$ (two upper panels) and the TE -statistic $TE(u)$ (two lower panels) as a function of the lower threshold u for positive returns of the DJIA at the daily time-scale. The two dashed lines of the left-panels show plus or minus one standard deviation (std) estimated as exposed in the text. The right-panels show the same statistics over the extended range of u .

Application to Indian Stock Index NSE

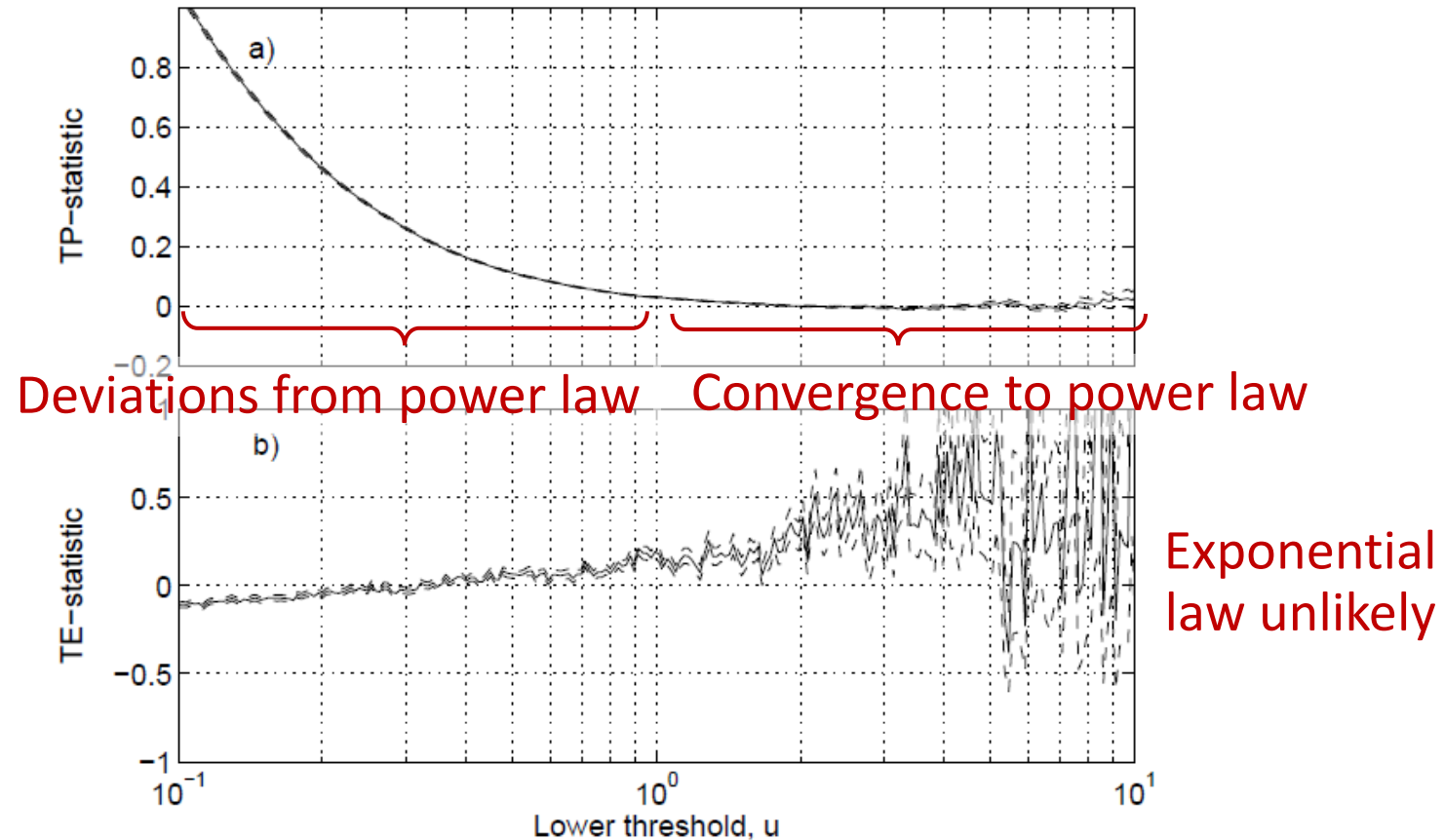


FIG. 2: (a) TP-statistic and (b) TE-statistic as function of the lower cut-off u for positive returns of the NSE with time interval $\Delta t = 1$ min. The broken lines indicate plus or minus one standard deviation of the statistics.

Remarks

- Graphical fitting of power laws are often subjective, particularly in the **choice of the lower cut-off**. This is illustrated by the different regimes of TP and TE statistics.
- TP and TE statistics confirm the tendency for the tails to resemble more and more a **power-law** for the highest quantiles.
- However, clear deviations suggest that the structure of the tails of the distributions of returns is **more complex** than usually assumed.
- Power laws are observed in **emerging markets** just like developed markets. Universal mechanisms?
- See Ref [6] for other simulated distributions (Student-t, stretched exponential, Gaussian). Their behaviors are different.
- See Ref [6] for the application to NASDAQ return distributions.

NSE Nifty Index (1 min return)

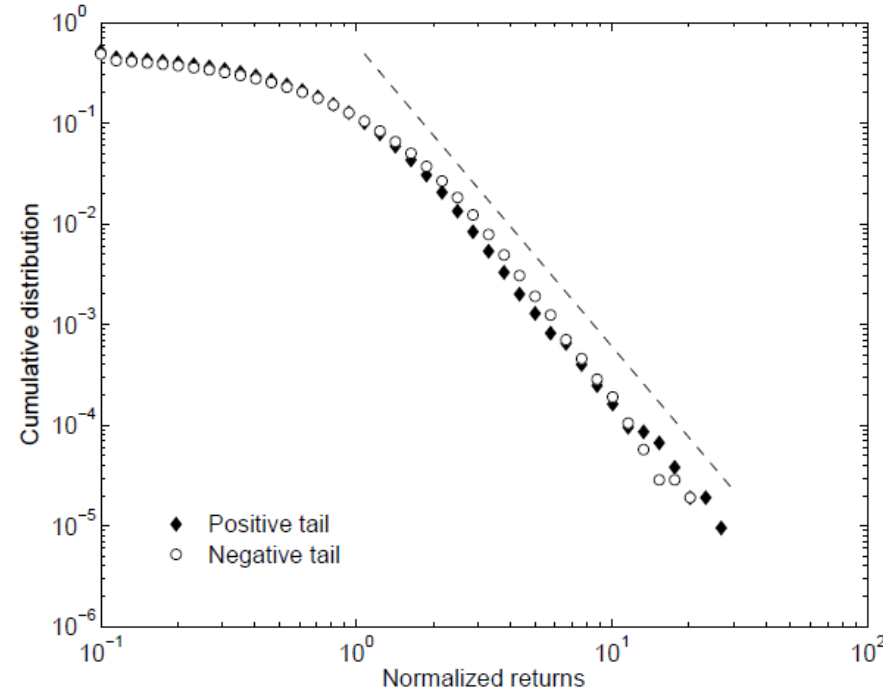


FIG. 3: The cumulative distribution of the normalized 1-min return for the NSE Nifty index. The broken line indicates a power-law with exponent $\alpha = 3$.

$$\alpha = \begin{cases} 2.98 \pm 0.09 & \text{(positive tail)} \\ 3.37 \pm 0.10 & \text{(negative tail)} \end{cases}$$

Confirmed the prediction of the TP statistic for threshold $> 10^0$.

Temporal Variations in the Return Distribution

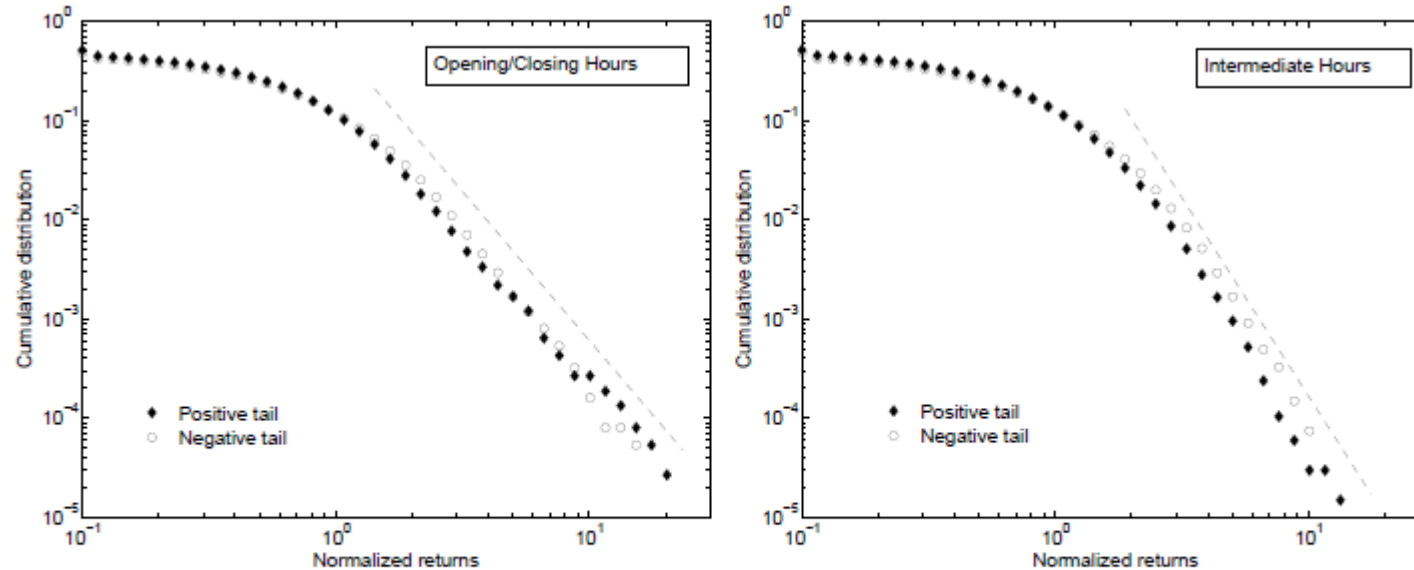


FIG. 4: Intra-day variation in the cumulative distribution of the normalized 1-min return for the NSE Nifty index: return distribution during (left) the opening and closing hours (the broken line indicates a power law with exponent $\alpha = 3$) and (right) the intermediate time period (the broken line indicates a power law with exponent $\alpha = 4$).

The average intra-day volatility of stock returns follows a U-shaped pattern.

This is reflected in the different exponents of the power laws for the opening/closing hours ($\alpha = 3$) and intermediate hours ($\alpha = 4$).

Dependence on Δt ?

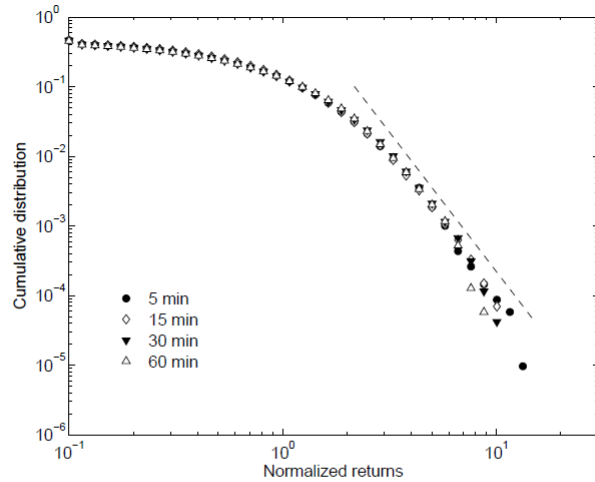


FIG. 5: The negative tail of the cumulative distribution of the NSE Nifty index returns for different time intervals Δt upto 60 min. The broken line indicates a power-law with exponent $\alpha = 4$.

Index	Δt	Power-law fit		Hill estimator	
		Positive	Negative	Positive	Negative
Nifty ('03-'04)	1 min	2.98 ± 0.09	3.37 ± 0.10	3.22 ± 0.03	3.47 ± 0.03
	5 min	4.42 ± 0.37	3.44 ± 0.21	4.51 ± 0.03	4.84 ± 0.03
	15 min	5.58 ± 0.88	3.96 ± 0.27	6.25 ± 0.03	4.13 ± 0.03
	30 min	5.13 ± 0.41	3.92 ± 0.45	5.65 ± 0.03	4.30 ± 0.03
	60 min	5.99 ± 1.52	4.42 ± 0.65	7.85 ± 0.03	5.11 ± 0.03
Nifty ('90-'06)	1 day	3.10 ± 0.34	3.18 ± 0.28	3.33 ± 0.14	3.37 ± 0.03
Sensex ('91-'06)	1 day	3.33 ± 0.77	3.45 ± 0.25	2.93 ± 0.15	3.84 ± 0.03

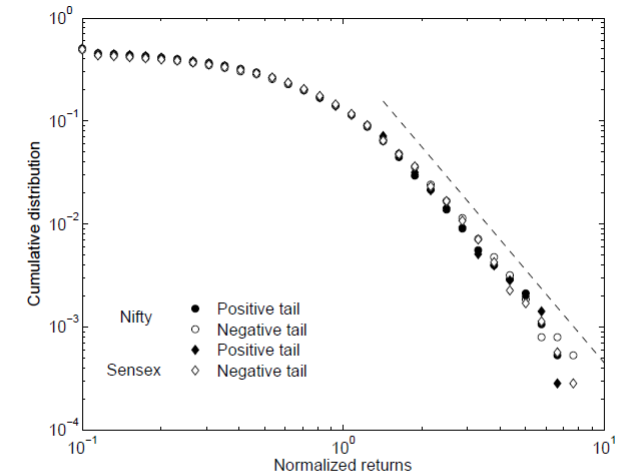


FIG. 6: The cumulative distribution of the normalized 1-day return for the NSE Nifty and BSE Sensex index. The broken line indicates a power-law with exponent $\alpha = 3$.

Index values are sampled at the end of intervals of duration Δt , or averaged over the interval. The distribution is a power law for Δt up to 60 minutes.

At longer time scales α increases and the distribution slowly converges to Gaussian behavior (not enough data!).

With enough data, $\Delta t = 1$ day still yields a power law with α close to 3.